

Bryan Zeng

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EXPERIENCE

INTEGRITY Security Services

Boston, MA

Full-Stack Software Engineer

January 2024 – December 2025

- Architected AWS-based full-stack web applications (Node.js, React) that support the **provisioning of 100,000+ IEEE 1609.2 certificates** for V2X infrastructure, streamlining the certificate management lifecycle
- Engineered cross-platform mobile apps (Kotlin, Swift) enabling **real-time interoperability** between roadside units, vehicles, and mobile devices, improving on-site hazard awareness and worker safety
- Implemented mobile signing libraries (Kotlin, Swift) **using ECDH key exchange** to enable secure, authenticated V2X messaging across transportation infrastructure
- Optimized system observability by **migrating 32M+ logs** to DocumentDB, accelerating queries and cutting overhead
- Containerized EV2G app and ensured **reliable CI/CD** by creating Jenkins pipelines for build, deployment, and testing
- Developed Flask microservice for **traffic intersection validation**, enabling Python-based validation from the web app
- Consolidated three repos into a monorepo to **facilitate end-to-end testing** and increase reusability and extensibility

Wolters Kluwer - UpToDate

Waltham, MA

Product Full-Stack Software Engineer

January 2023 – December 2023

- Spearheaded modernization initiatives (Java, Spring Boot, Vue.js) **eliminating technical debt** and resolving critical security vulnerabilities for a medical resources platform **serving 2M+ health professionals**
- Enhanced internal developer tooling for **150+ developers** to ensure proper backend configuration, reducing setup errors and accelerating onboarding
- Redesigned error-handling front-end architecture across application, improving user experience and maintainability

NU Rover

Boston, MA

Software Developer | Webmaster

September 2021 – May 2025

- Co-developed a **small-scale haptic arm** in C++ as a PoC of controlling a full 4-degree-of-freedom robotic arm
- Built **Python ROS state machine**, powering autonomous Rover navigation via Zed2 camera GPS and IMU data

PROJECTS

Large-Scale Misinformation Classification

September 2025 – Present

- Trained deep learning models (**CNN, LSTM, FNN, Longformer**) for misinformation detection across **3.4M articles**, achieving **99.6% accuracy** with Longformer outperforming baseline SVM/XGBoost models

Cheffy: AI Cooking Assistant

September 2025 – Present

- Co-designed AI agentic search cooking assistant using **Gemini and multi-agent RAG** to autonomously retrieve recipes and provide personalized cooking guidance

Finance Tracker AI Chatbot

June 2025 – Present

- Created RAG-powered chatbot using LangChain for automated financial analysis transaction categorization and personalized budget insights

Twitter Sentiment Analysis Stock Portfolio

October 2022 – December 2022

- Constructed quantitative trading model using NLP sentiment analysis (NLTK, Scikit-Learn) on **1.75M+ tweets**

EDUCATION

Northeastern University - Khoury College of Computer Sciences

Boston, MA

M.S. in Computer Science | GPA: 4.0/4.0

Expected May 2026

B.S. in Computer Science and Business Administration | GPA: 3.966/4.0

Key Coursework: Artificial Intelligence, Machine Learning, Software Engineering, Networks, Database Management, Systems, Algorithms & Data, Object-Oriented Design, Data Science

TECHNICAL SKILLS

- **Languages:** JavaScript/TypeScript, Java, Python, Kotlin, Swift, Go, SQL, C++
- **Frameworks:** React, Node.js, Vue, Spring Boot, Flask
- **AI/ML Tools:** PyTorch, TensorFlow, Scikit-Learn, LangChain, Ollama
- **Technologies:** AWS (CloudFormation, EC2, S3, Lambda), Docker, Jenkins, Git, Jira, MongoDB